cover">

AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 2 Differentiation

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings)
- Questions 91–100: Free Response with Official AP Scoring Rubrics
- Complete Answer Key and Detailed Worked Solutions at the End

OFFICIAL AP CED ALIGNED · MIXED LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This paper contains 100 continuously numbered questions. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★	**Very Important**	Core AP exam concepts: Limit definition of derivative, Chain Rule, Implicit Differentiation, Related Rates setup, and f' vs f'' graphical behavior.
★★★	**Mandatory**	Essential skills: Product & Quotient Rules, trig/exponential derivatives, tangent and normal line equations, and average rate of change.
★★	**Important**	Foundational skills: Power rule, constant functions, simple polynomials, and basic graphical rates of change.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer.
- **Questions 91–100:** Free response, 9 points each, with analytical setups and rubrics.
- **Difficulty Mix:** Easy 5 · Moderate 68 · Hard 17.
- **Calculator Policy:** No calculator required.
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 2.1 – 2.10)

2.1 Defining Average & Instantaneous Rate

2.2 Defining the Derivative

2.3 Estimating Derivatives

2.4 Differentiability & Continuity

2.5 Power Rule

2.6 Constant, Sum, Difference & Constant Multiple

2.7 Derivatives of cos, sin, e^x, ln x

2.8 Product Rule

2.9 Quotient Rule

2.10 Chain Rule

2.11 Implicit Differentiation

2.12 Higher Order Derivatives

> **Study Tip:** Master the Chain Rule and Implicit Differentiation (★★★★★) before moving forward into applications!

ITEM 01

SINGLE BEST CHOICE

★★ IMPORTANT

The average rate of change of $f(x) = x^2 + 3x$ on the interval $[1, 4]$ is

A 8

B 5

C 7

D 3

ITEM 02

SINGLE BEST CHOICE

★★ IMPORTANT

If $f(x) = 5$, what is the average rate of change of f on any interval $[a, b]$?

A Undefined

B 5

C $b - a$

D 0

ITEM 03

SINGLE BEST CHOICE

★★★ MANDATORY

The difference quotient $[f(x+h) - f(x)]/h$ for $f(x) = x^2$ simplifies to

A h

B $2x - h$

C $2x$

D $2x + h$

ITEM 04

SINGLE BEST CHOICE

★★★ MANDATORY

Let $f(x) = \sqrt{x}$. What is the average rate of change on $[4, 9]$?

A 5

B 1

C 1/3

D 1/5

ITEM 05

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Which expression represents the instantaneous rate of change of f at $x = a$?

A $[f(a+h) - f(a)]/h$

B $\lim_{x \to a} f(x)$

C $f(a+h) - f(a)$

D $\lim_{h \to 0} [f(a+h) - f(a)]/h$

ITEM 06

SINGLE BEST CHOICE

★★ IMPORTANT

If $f(x) = 3x - 7$, then $f'(2) =$

A −7

B 0

C 3

D 2

ITEM 07

SINGLE BEST CHOICE

★★★ MANDATORY

The graph of f passes through (2, 5) and (4, 11). Estimate f′(3).

A 5

B 2

C 6

D 3

ITEM 08

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = x^3$, what is $\lim_{h \to 0} [f(2+h) - f(2)]/h$?

A 12

B 8

C 0

D 4

ITEM 09

SINGLE BEST CHOICE

★★★ MANDATORY

A particle moves such that $s(t) = t^2 + 2t$. Its average velocity on $[1, 3]$ is

A 10 m/s

B 8 m/s

C 4 m/s

D 6 m/s

ITEM 10

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f is differentiable at $x = c$, then f must be

A Linear near $x = c$

B Defined at $x = c$ only

C Constant near $x = c$

D Continuous at $x = c$

ITEM 11

SINGLE BEST CHOICE

★★ IMPORTANT

If $f(x) = x^5$, then $f'(x) =$

A x^4

B $5x$

C $5x^5$

D $5x^4$

ITEM 12

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $g(t) = 4t^3 - 2t + 7$ is

A $12t^3 - 2$

B $12t^2 - 2t$

C $4t^2 - 2$

D $12t^2 - 2$

ITEM 13

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $y = \sqrt{x}$, then $dy/dx =$

A $1/(2\sqrt{x})$

B $2\sqrt{x}$

C $1/\sqrt{x}$

D $-1/(2\sqrt{x})$

ITEM 14

SINGLE BEST CHOICE

★★★ MANDATORY

The slope of the tangent line to $y = x^2$ at $x = -1$ is

A 2

B −1

C −2

D 1

ITEM 15

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = 1/x$, then $f'(2) =$

A −1/4

B −1/2

C 1/4

D 1/2

ITEM 16

SINGLE BEST CHOICE

★★ IMPORTANT

Which function has a derivative equal to 0 for all x?

A f(x) = x

B f(x) = sin x

C $f(x) = x^2$

D f(x) = 5

ITEM 17

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = x^n$, and $f'(1) = 10$, what is n?

A 1

B 10

C 5

D 0

ITEM 18

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $h(x) = \pi^3$ is

A $3\pi^2$

B π^2

C Undefined

D 0

ITEM 19

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = 2x^4 - 3x^2$, then $f'(-1) =$

A −2

B −14

C 2

D 14

ITEM 20

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The equation of the tangent line to $y = x^3$ at $x = 1$ is

A y = x − 2

B y = 3x − 2

C y = 3x

D y = 3x + 2

ITEM 21

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = ax^2 + bx$, and $f'(1) = 4$, $f'(2) = 7$, find a.

A 3/2

B 1

C 2

D 1/2

ITEM 22

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = x^2 \sin x$, then $f'(x) =$

A $2x \sin x + x^2 \cos x$

B $2x \sin x - x^2 \cos x$

C $2x \cos x$

D $x^2 \cos x$

ITEM 23

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $g(x) = e^x / x$ is

A $e^x(x+1)/x^2$

B $(xe^x - e^x)/x^2$

C e^x/x

D $(e^x - xe^x)/x^2$

ITEM 24

SINGLE BEST CHOICE

★★★ MANDATORY

If $h(x) = (x^2 + 1)(x)$, then $h'(2) =$

A 12

B 15

C 10

D 13

ITEM 25

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $f(x) = \tan x$ is

A $-\cot x$

B $\sec^2 x$

C $-\csc^2 x$

D $\cot x$

ITEM 26

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \cos x$, then $f'(\pi/2) =$

A -1

B $\pi/2$

C 0

D 1

ITEM 27

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $y = x \sin x + \cos x$ is

A $\sin x + x \cos x - \sin x$

B $x \cos x$

C $\cos x$

D $x \cos x - \sin x$

ITEM 28

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = e^x \cos x$, then $f'(0) =$

A 2

B 1

C −1

D 0

ITEM 29

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The slope of the normal line to $y = x^2$ at $x = 1$ is

A 2

B −2

C −1/2

D 1/2

ITEM 30

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = (x^2 - 1)/(x + 1)$, then $f'(2) =$

A 2

B 1

C 4

D 3

ITEM 31

SINGLE BEST CHOICE

★★★ MANDATORY

Which is the derivative of $f(x) = x^5 e^x$?

A $x^4 e^x (5 + x)$

B $x^5 e^x$

C $5x^4 + e^x$

D $5x^4 e^x$

ITEM 32

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = \sin x / x$, then $f'(\pi) =$

A $-1/\pi$

B $-1/\pi^2$

C 0

D $1/\pi$

ITEM 33

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $f(x) = 3^x$ is

A 3^x

B $3^x \ln 3$

C $x\, 3^{x-1}$

D $\ln 3$

ITEM 34

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \ln x$, then $f'(e) =$

A 0

B 1

C e

D 1/e

ITEM 35

SINGLE BEST CHOICE

★★★ MANDATORY

The tangent line to $y = e^x$ at $x = 0$ passes through

A (1, 0)

B (1, 1)

C (0, 0)

D (0, 1)

ITEM 36

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = x^2 + 2x$, where is the tangent horizontal?

A $x = 1$

B $x = -2$

C $x = -1$

D $x = 0$

ITEM 37

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $f(x) = \cot x$ is

A $\sec^2 x$

B $-\sec^2 x$

C $-\csc^2 x$

D $\csc^2 x$

ITEM 38

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \sqrt{x} \cdot x$, then $f'(4) =$

A 3/4

B 6

C 3

D 3/2

ITEM 39

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The normal line to $y = \sin x$ at $x = \pi/2$ is

A $y = x$

B $x = \pi/2$

C $y = 0$

D $y = 1$

ITEM 40

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = e^{(2x)}$, then $f'(0) =$

A 1

B 2

C e

D 2e

ITEM 41

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $y = \ln(2x)$ is

A $1/(2x)$

B $1/x$

C $\ln 2$

D $2/x$

ITEM 42

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = (x^2 + 1)^5$, then $f'(x) =$

A $10x(x^2 + 1)^4$

B $5(x^2 + 1)^4 \cdot 2x$

C $5(x^2 + 1)^4$

D $10x(x^2 + 1)^5$

ITEM 43

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $g(x) = \sin(x^2)$ is

A $-\cos(x^2)$

B $\cos(x^2)$

C $2x\cos(x^2)$

D $2x\sin(x^2)$

ITEM 44

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $h(x) = e^{\wedge}(x^2)$, then $h'(1) =$

A e^2

B $2e$

C e

D $2e^2$

ITEM 45

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $f(x) = \sqrt{(3x + 1)}$ is

A $1/\sqrt{(3x+1)}$

B $3/(2\sqrt{(3x+1)})$

C $1/(2\sqrt{(3x+1)})$

D $3/\sqrt{(3x+1)}$

ITEM 46

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = (\sin x)^3$, then $f'(\pi/2) =$

A −3

B 1

C 3

D 0

ITEM 47

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $y = \ln(\cos x)$ is

A sec x

B −sec x

C tan x

D −tan x

ITEM 48

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f(x) = e^(sin x), then f'(0) =

A 0

B sin 1

C 1

D e

ITEM 49

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $f(x) = (x + \sqrt{x})^2$ is

A $2(x + \sqrt{x})(1 + 1/(2\sqrt{x}))$

B $2x + 2\sqrt{x}$

C $(1 + 1/(2\sqrt{x}))^2$

D $2(x + \sqrt{x})$

ITEM 50

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \cos(x^3)$, then $f'(x) =$

A $-3x^2 \sin(x^3)$

B $\sin(3x^2)$

C $3x^2 \sin(x^3)$

D $-\sin(x^3)$

ITEM 51

SINGLE BEST CHOICE

★★★ MANDATORY

The slope of the tangent line to $y = (2x - 1)^4$ at $x = 1$ is

A 32

B 16

C 8

D 4

ITEM 52

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f(x) = ln(ln x), then f'(e) =

A 1/e

B 1

C 1/(e ln e)

D e

ITEM 53

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of y = e^(e^x) is

A e^x · e^(e^x) · e^x

B e^(e^x) · e^x

C e^(e^x)

D e^(e^x + x)

ITEM 54

SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = √(x² + 1), then f'(0) =

A 1

B 0

C Undefined

D 1/2

ITEM 55

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $f(x) = \sin^2 x + \cos^2 x$ is

A $-2 \sin x \cos x$

B $2 \sin x \cos x$

C 1

D 0

ITEM 56

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = (x^2 - 1)^{100}$, then $f'(1) =$

A 200

B 100

C 1

D 0

ITEM 57

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The derivative of $y = \tan(\sqrt{x})$ is

A $\sec^2(\sqrt{x})$

B $\sec(\sqrt{x}) \tan(\sqrt{x})/(2\sqrt{x})$

C $1/(2\sqrt{x})$

D $\sec^2(\sqrt{x})/(2\sqrt{x})$

ITEM 58

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = e^{(-x^2)}$, where is $f'(x) = 0$?

A $x = -1$

B Nowhere

C $x = 0$

D $x = 1$

ITEM 59

SINGLE BEST CHOICE

★★★ MANDATORY

The derivative of $f(x) = \ln(x^2)$ for $x > 0$ is

A $2/x$

B $2x$

C $1/x^2$

D $1/(2x)$

ITEM 60

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^2 + y^2 = 25$, then $dy/dx =$

A $-x/y$

B x/y

C $-y/x$

D y/x

ITEM 61

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

For the curve $xy = 1$, $dy/dx =$

A $-y/x$

B y/x

C $-x/y$

D x/y

ITEM 62

SINGLE BEST CHOICE

★★★ MANDATORY

If $x^2 + y^2 = 100$, the slope of the curve at $(6, 8)$ is

A $4/3$

B $-4/3$

C $-3/4$

D $3/4$

ITEM 63

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Implicit differentiation of $\sin(xy) = x$ gives

A $\cos(xy)(xy') = 1$

B $y\cos(xy) + xy' = 1$

C $\cos(xy) = 1$

D $\cos(xy)(y + xy') = 1$

ITEM 64

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^3 + y^3 = 6xy$, then dy/dx at $(3, 3)$ is

A 1

B Undefined

C 0

D −1

ITEM 65

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

A 10-foot ladder slides down a wall. If x is the distance from the wall and y is the height on the wall, the relation is

A $x^2 + y^2 = 100$

B $xy = 10$

C $x + y = 10$

D $x^2 - y^2 = 100$

ITEM 66

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If the radius of a circle increases at 2 cm/s, how fast does its area increase when $r = 3$ cm?

A 3π cm²/s

B 4π cm²/s

C 6π cm²/s

D 12π cm²/s

ITEM 67

SINGLE BEST CHOICE

★★★ MANDATORY

For the hyperbola $x^2 - y^2 = 1$, $dy/dx =$

A $-x/y$

B x/y

C $-y/x$

D y/x

ITEM 68

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $y = f(u)$ and $u = g(t)$, then $dy/dt =$

A $f'(g(t))\, g'(t)$

B $f(g'(t))$

C $f'(u)\, g'(t)$

D $g'(f(u))$

ITEM 69

SINGLE BEST CHOICE

★★★ MANDATORY

The tangent line to the circle $x^2 + y^2 = 25$ is vertical when

A $x = 0$

B $x = 5$

C $y = 0$

D $y = 5$

ITEM 70

SINGLE BEST CHOICE

★★ IMPORTANT

If $2x + 3y = 6$, then $dy/dx =$

A $-2/3$

B $2/3$

C $3/2$

D $-3/2$

ITEM 71

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Implicit differentiation of e^(xy) = x + y gives

A x e^(xy) = 1

B e^(xy)(xy') = 1 + y'

C y e^(xy) = 1

D e^(xy)(y + xy') = 1 + y'

ITEM 72

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x = \cos t$ and $y = \sin t$, then $dy/dx =$

A $-\tan t$

B $-\cot t$

C $\tan t$

D $\cot t$

ITEM 73

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The volume of a sphere is $V = (4/3)\pi r^3$. If $dr/dt = k$, then $dV/dt =$

A $(4/3)\pi r^2 k$

B $\pi r^2 k$

C $4\pi r^2 k$

D $4\pi r k$

ITEM 74

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $y^2 = x$, then $d^2y/dx^2 =$

A $-1/(2y^3)$

B $-1/(4y^3)$

C $1/(4y^3)$

D $1/(2y)$

ITEM 75

SINGLE BEST CHOICE

★★★ MANDATORY

The normal line to $xy = 4$ at the point $(2, 2)$ has slope

A 2

B -1

C -2

D 1

ITEM 76

SINGLE BEST CHOICE

★★ IMPORTANT

If $f'(x) > 0$ on (a, b), then f is

A Increasing

B Concave Down

C Concave Up

D Decreasing

ITEM 77

SINGLE BEST CHOICE

★★ IMPORTANT

If $f''(x) < 0$ on (a, b), then the graph of f is

A Decreasing

B Increasing

C Concave Down

D Concave Up

ITEM 78

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

A point of inflection on a continuous curve occurs where

A $f'(x)$ is maximized

B $f'(x) = 0$

C $f(x) = 0$

D $f''(x) = 0$ or undefined, and concavity changes sign

ITEM 79

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f'(c) = 0$ and $f''(c) > 0$, then f has a

A Point of Inflection at c

B None of the above

C Local Maximum at c

D Local Minimum at c

ITEM 80

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Given the graph of f', a local maximum of f occurs where

A f' has a maximum

B f' has a minimum

C f' crosses the x-axis from positive to negative

D f' crosses the x-axis from negative to positive

ITEM 81

SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = x^4$, then $f''(0) =$

A 12

B 4

C 24

D 0

ITEM 82

SINGLE BEST CHOICE

★★ IMPORTANT

Velocity v(t) is defined as the first derivative of

A Acceleration a(t)

B Time t

C Position s(t)

D Speed |v(t)|

ITEM 83

SINGLE BEST CHOICE

★★ IMPORTANT

Acceleration a(t) is defined as the first derivative of

A Position s(t)

B Displacement

C Speed |v(t)|

D Velocity v(t)

ITEM 84

SINGLE BEST CHOICE

★★★ MANDATORY

If $f'(x) = x(x - 2)(x + 1)$, what are the critical numbers of f?

A 0, −2, 1

B 0, 2, −1

C 0, 1, 2

D 1, −1, 2

ITEM 85

SINGLE BEST CHOICE

★★★ MANDATORY

If $f''(x) = 6x - 6$, where does the graph of f have a point of inflection?

A $x = -1$

B $x = 0$

C $x = 6$

D $x = 1$

ITEM 86

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Values of differentiable functions f and g are given: $f(2) = 3$, $f'(2) = 5$, $g(2) = -1$, $g'(2) = 4$. What is $(f \cdot g)'(2)$?

A 20

B −5

C 17

D 7

ITEM 87

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Where is the function $f(x) = |x - 3|$ not differentiable?

A $x = -3$

B f is differentiable everywhere

C $x = 0$

D $x = 3$

ITEM 88

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

For $f(x) = x / (x^2 + 1)$, at what values of x does the graph have a horizontal tangent line?

A $x = 1$ and $x = -1$

B There are no horizontal tangents

C $x = 1$ only

D $x = 0$ only

ITEM 89

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = \sec x$, what is $f''(0)$?

A 0

B 1

C 2

D −1

ITEM 90

SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = x^2 - 2x$, on the interval $[0, 4]$, at what value of c does the instantaneous rate of change equal the average rate of change?

A 2

B 1

C 3

D 4

Question 91: Difference Quotient & Tangent Line

★★★★★ VERY IMPORTANT

9 Points

Let f be a function defined by $f(x) = x^2 - 4x + 5$.

(a) [2 Points] Find the average rate of change of f on the interval [1, 3].

(b) [4 Points] Using the limit definition of the derivative, find f'(x). Show all steps.

(c) [3 Points] Write an equation for the line tangent to the graph of f at x = 2.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 92: Product and Quotient Rules Application

★★★★★ VERY IMPORTANT | 9 Points

Let $f(x) = x^2 e^x$ and $g(x) = (x + 1)/(x - 1)$.

(a) [3 Points] Find f'(x) using the Product Rule.

(b) [3 Points] Find g'(x) using the Quotient Rule.

(c) [3 Points] Evaluate f'(0) and g'(2).

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 93: Chain Rule with Trig and Exponentials

★★★★★ VERY IMPORTANT | 9 Points

Let $h(x) = \sin(e^{(2x)})$ and $k(x) = \ln(\cos x)$.

(a) [3 Points] Find $h'(x)$.

(b) [3 Points] Find $k'(x)$.

(c) [3 Points] Evaluate $h'(0)$ and $k'(\pi/4)$.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 94: Implicit Differentiation & Tangent Line

★★★★★ VERY IMPORTANT | 9 Points

Consider the curve defined by $x^2 + xy + y^2 = 3$.

(a) [3 Points] Show that $dy/dx = -(2x + y)/(x + 2y)$.

(b) [2 Points] Find the coordinates of all points on the curve where $x = 1$.

(c) [4 Points] Write an equation for the line tangent to the curve at the point (1, 1).

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 95: Related Rates: Ladder Problem

★★★★★ VERY IMPORTANT

9 Points

A 10-foot ladder leans against a vertical wall. The bottom of the ladder slides away from the wall at a rate of 2 ft/sec.

(a) [2 Points] Write an equation relating the distance x from the wall to the height y on the wall.

(b) [4 Points] Find dy/dt when x = 6 feet.

(c) [3 Points] Is the top of the ladder moving up or down at this instant? Justify.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 96: Higher Order Derivatives & Concavity

★★★★★ VERY IMPORTANT | 9 Points

Let $f(x) = x^4 - 4x^3$.

(a) [2 Points] Find $f'(x)$ and $f''(x)$.

(b) [3 Points] Find the x-coordinates of all points of inflection.

(c) [4 Points] Determine the intervals where the graph of f is concave up.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 97: Graphical Analysis of Derivative f'

★★★★★ VERY IMPORTANT

9 Points

The continuous derivative f' is positive on $(-2, 1)$ and $(4, \infty)$, and negative on $(-\infty, -2)$ and $(1, 4)$.

(a) [2 Points] On what intervals is f increasing?

(b) [3 Points] At what x-values does f have a local maximum?

(c) [4 Points] If f(0) = 5, estimate f(2) given that the net change $\int_0^2 f'(x)\,dx$ is −3.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 98: Parametric Derivatives & Motion

★★★★★ VERY IMPORTANT

9 Points

A particle moves in the xy-plane with position $(x(t), y(t))$ given by $x(t) = t^2 - 1$ and $y(t) = t^3 - 3t$.

(a) [3 Points] Find dy/dx in terms of t.

(b) [2 Points] Find the slope of the path at $t = 2$.

(c) [4 Points] For what values of t is the tangent line horizontal?

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 99: Tangent Line & Local Linear Approximation

★★★★★ VERY IMPORTANT 9 Points

Let f be a differentiable function such that $f(4) = 2$ and $f'(4) = -3$.

(a) [2 Points] Write an equation for the line tangent to the graph of f at x = 4.

(b) [3 Points] Use the tangent line to approximate f(4.1).

(c) [4 Points] If f''(x) < 0 for all x, is the approximation in part (b) an overestimate or underestimate? Justify.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 100: Second Derivative Test & Critical Values

★★★★★ VERY IMPORTANT | 9 Points

Let $f(x) = x + 2\sin x$ on the closed interval $[0, 2\pi]$.

(a) [2 Points] Find $f'(x)$ and $f''(x)$.

(b) [3 Points] Find the critical numbers of f in the interval.

(c) [4 Points] Use the Second Derivative Test to classify each critical number as a local maximum or local minimum.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point rubrics.
- **Appendix:** Essential Unit 2 Derivative Rules & Formulas formulas and theorems.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
A ★★	D ★★	D ★★★	D ★★★	D ★★★★★	C ★★	D ★★★	A ★★★★★	D ★★★	D ★★★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
D ★★	D ★★★	A ★★★★★	C ★★★	A ★★★	D ★★	B ★★★	D ★★★	A ★★★	B ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
A ★★★★★	A ★★★★★	B ★★★★★	D ★★★	B ★★★★★	A ★★★	B ★★★★★	B ★★★★★	C ★★★★★	B ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
A ★★★	A ★★★★★	B ★★★★★	D ★★★	D ★★★	C ★★★	C ★★★	C ★★★	B ★★★★★	B ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
B ★★★★★	A ★★★★★	C ★★★★★	B ★★★★★	B ★★★	D ★★★★★	D ★★★★★	C ★★★★★	A ★★★	A ★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★★	A ★★★★★	B ★★★★★	B ★★★	D ★★★★★	D ★★★	D ★★★★★	C ★★★★★	A ★★★	A ★★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
A ★★★★★	C ★★★	D ★★★★★	D ★★★★★	A ★★★★★	D ★★★★★	B ★★★	A ★★★★★	C ★★★	A ★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
D ★★★★★	B ★★★★★	C ★★★★★	B ★★★★★	D ★★★	A ★★	C ★★	D ★★★★★	D ★★★★★	C ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
D ★★★	C ★★	D ★★	B ★★★	D ★★★	D ★★★★★	D ★★★★★	A ★★★★★	B ★★★★★	A ★★★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01 Analysis

★★ IMPORTANT

Correct Answer: (A) 8

ANALYTICAL STEP-BY-STEP SOLUTION

Average Rate = [f(4) − f(1)] / (4 − 1). f(4) = 16 + 12 = 28. f(1) = 1 + 3 = 4. (28 − 4)/3 = 24/3 = 8.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates slope between (1,1) and (4,4) incorrectly.
✗ **(C)** Arithmetic error in subtraction (28 − 4 = 21).
✗ **(D)** Uses f(4) − f(1) without dividing by 3.

Question 02 Analysis

★★ IMPORTANT

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

f is a constant function. The change in y is always 0. Thus, average rate = 0/(b − a) = 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Division by zero only happens if a = b, but intervals imply a ≠ b.
✗ **(B)** Confuses the function value with the rate of change.
✗ **(C)** Confuses the denominator with the rate.

Question 03 Analysis

★★★ MANDATORY

Correct Answer: (D) 2x + h

ANALYTICAL STEP-BY-STEP SOLUTION

$[(x+h)^2 - x^2]/h = [x^2 + 2xh + h^2 - x^2]/h = (2xh + h^2)/h = h(2x+h)/h = 2x + h$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Only keeps the h^2 term.
✗ **(B)** Sign error in expansion.
✗ **(C)** Takes the limit as $h \to 0$ immediately.

Question 04 Analysis

★★★ MANDATORY

Correct Answer: (D) 1/5

ANALYTICAL STEP-BY-STEP SOLUTION

[f(9) − f(4)] / (9 − 4) = (3 − 2) / 5 = 1/5.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds instead of subtracts.
✗ **(B)** Uses (9 − 4)/(3 − 2) = 5.
✗ **(C)** Uses 1/(√9 − √4) = 1/1 = 1.

Question 05 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) *lim* (h→0) [f(a+h) − f(a)]/h

ANALYTICAL STEP-BY-STEP SOLUTION

Definition of derivative: f'(a) = ***lim*** (h→0) [f(a+h) − f(a)]/h.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the difference quotient, not the limit.
✗ **(B)** This is the limit of the function, not the rate.
✗ **(C)** This is just the change in y.

Question 06 Analysis

★★ IMPORTANT

Correct Answer: (C) 3

ANALYTICAL STEP-BY-STEP SOLUTION

f is linear with slope 3. The derivative of a linear function is its constant slope, so f'(2) = 3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses y-intercept with slope.
✗ **(B)** Evaluates f(2) = −1 and guesses 0.
✗ **(D)** Confuses input x = 2 with output.

Question 07 Analysis

★★★ MANDATORY

Correct Answer: (D) 3

ANALYTICAL STEP-BY-STEP SOLUTION

Approximate derivative using the secant slope: (11 − 5) / (4 − 2) = 6 / 2 = 3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses average of y-values.
✗ **(B)** Uses only y-values difference.
✗ **(C)** Uses product of differences.

Question 08 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 12

ANALYTICAL STEP-BY-STEP SOLUTION

This limit is the definition of f'(2). f'(x) = $3x^2$. f'(2) = $3(2)^2$ = 12.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates 2^3 = 8.
✗ **(C)** Thinks derivative of cubic is 0 at some point.
✗ **(D)** Calculates 2^2 = 4.

Question 09 Analysis

★★★ MANDATORY

Correct Answer: (D) 6 m/s

ANALYTICAL STEP-BY-STEP SOLUTION

[s(3) − s(1)] / (3 − 1). s(3) = 9 + 6 = 15. s(1) = 1 + 2 = 3. (15 − 3)/2 = 12/2 = 6 m/s.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds s(3) + s(1).
✗ **(B)** Forgets to divide by time interval (15 − 3 = 12).
✗ **(C)** Calculates s(3) only.

Question 10 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) Continuous at x = c

ANALYTICAL STEP-BY-STEP SOLUTION

Differentiability implies continuity. If a function has a derivative at a point, it must be continuous there.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Not necessarily linear (e.g. x^2).

✗ **(B)** It must be defined, but continuity is the stronger necessary condition.

✗ **(C)** Not necessarily constant.

Question 11 Analysis

★★ IMPORTANT

Correct Answer: (D) $5x^4$

ANALYTICAL STEP-BY-STEP SOLUTION

Power Rule: $d/dx\ (x^n) = nx^{n-1}$. So $5x^4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the coefficient n.

✗ **(B)** Subtracts 1 from coefficient.

✗ **(C)** Forgets to subtract 1 from exponent.

Question 12 Analysis

★★★ MANDATORY

Correct Answer: (D) $12t^2 - 2$

ANALYTICAL STEP-BY-STEP SOLUTION

$d/dt(4t^3) = 12t^2$. $d/dt(-2t) = -2$. $d/dt(7) = 0$. Result: $12t^2 - 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doesn't reduce power on first term.

✗ **(B)** Keeps the t in the second term.

✗ **(C)** Differentiates $4t^3$ as $4t^2$.

Question 13 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1/(2√x)

ANALYTICAL STEP-BY-STEP SOLUTION

√x = x^(1/2). Derivative is (1/2)x^(−1/2) = 1/(2√x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Multiplies by 2 instead of 1/2.
✗ **(C)** Forgets the 1/2 factor.
✗ **(D)** Sign error.

Question 14 Analysis

★★★ MANDATORY

Correct Answer: (C) −2

ANALYTICAL STEP-BY-STEP SOLUTION

y' = 2x. At x = −1, slope = 2(−1) = −2.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Squares the x-value again.
✗ **(B)** Evaluates y instead of y'.
✗ **(D)** Sign error (thinks slope is positive).

Question 15 Analysis

★★★ MANDATORY

Correct Answer: (A) −1/4

ANALYTICAL STEP-BY-STEP SOLUTION

$f(x) = x^{-1}$. $f'(x) = -x^{-2} = -1/x^2$. $f'(2) = -1/4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the negative sign.
✗ **(C)** Sign error and square error.
✗ **(D)** Positive reciprocal.

Question 16 Analysis

★★ IMPORTANT

Correct Answer: (D) f(x) = 5

ANALYTICAL STEP-BY-STEP SOLUTION

The derivative of any constant function is 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Derivative is 1.
✗ **(B)** Derivative is cos x.
✗ **(C)** Derivative is 2x.

Question 17 Analysis

★★★ MANDATORY

Correct Answer: (B) 10

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = nx^{n-1}$. $f'(1) = n(1)^{n-1} = n$. Given $f'(1) = 10$, so $n = 10$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Thinks n must be 1.
✗ **(C)** Thinks n must be half.
✗ **(D)** Thinks n must be 0.

Question 18 Analysis

★★★ MANDATORY

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

π is a constant, so π^3 is also a constant. The derivative of any constant is 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats π as a variable x.
✗ **(B)** Treats π as variable without power rule.
✗ **(C)** Constants are differentiable.

Question 19 Analysis

★★★ MANDATORY

Correct Answer: (A) −2

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = 8x^3 - 6x$. $f'(-1) = 8(-1) - 6(-1) = -8 + 6 = -2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates $8(-1) - 6 = -14$ (sign error on second term).
✗ **(C)** Sign error on first term.
✗ **(D)** Adds magnitudes.

Question 20 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) y = 3x – 2

ANALYTICAL STEP-BY-STEP SOLUTION

Point: (1, 1). Slope: $y' = 3x^2 \rightarrow 3(1)^2 = 3$. Eq: $y - 1 = 3(x - 1) \rightarrow y = 3x - 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Wrong slope (1).
✗ **(C)** Passes through origin, wrong intercept.
✗ **(D)** Wrong intercept (+2).

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 3/2

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = 2ax + b$. 1) $2a + b = 4$. 2) $4a + b = 7$. Subtract (1) from (2): $2a = 3 \rightarrow a = 3/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Assumes $a = 1$.
✗ **(C)** Arithmetic error $2a = 4$.
✗ **(D)** Solves for b instead.

Question 22 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $2x \sin x + x^2 \cos x$

ANALYTICAL STEP-BY-STEP SOLUTION

Product Rule: u'v + uv'. u = x^2, v = sin x. u' = 2x, v' = cos x. Result: $2x \sin x + x^2 \cos x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error (quotient rule confusion).
✗ **(C)** Derivative of product is NOT product of derivatives.
✗ **(D)** Misses the first term (u'v).

Question 23 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $(xe^x - e^x)/x^2$

ANALYTICAL STEP-BY-STEP SOLUTION

Quotient Rule: $(u'v - uv')/v^2$. u = e^x, v = x. $(e^x \cdot x - e^x \cdot 1)/x^2 = (xe^x - e^x)/x^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error in numerator addition.
✗ **(C)** Forgot quotient rule entirely.
✗ **(D)** Numerator terms swapped (sign error).

Question 24 Analysis

★★★ MANDATORY

Correct Answer: (D) 13

ANALYTICAL STEP-BY-STEP SOLUTION

$h'(x) = (2x)(x) + (x^2+1)(1) = 2x^2 + x^2 + 1 = 3x^2 + 1$. $h'(2) = 3(4) + 1 = 13$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates $3x^2$ only.
✗ **(B)** Arithmetic error.
✗ **(C)** Calculates $2x^2$ only.

Question 25 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\sec^2 x$

ANALYTICAL STEP-BY-STEP SOLUTION

Standard derivative: d/dx (tan x) = $\sec^2 x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Negative reciprocal.
✗ **(C)** Derivative of cot x.
✗ **(D)** Reciprocal function.

Question 26 Analysis

★★★ MANDATORY

Correct Answer: (A) −1

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = −sin x. f'(π/2) = −sin(π/2) = −1.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Confuses input with output.
✗ **(C)** Evaluates cos instead of sin.
✗ **(D)** Sign error.

Question 27 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) x cos x

ANALYTICAL STEP-BY-STEP SOLUTION

d/dx (x sin x) = 1·sin x + x·cos x. d/dx (cos x) = −sin x. Sum: sin x + x cos x − sin x = x cos x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot to cancel sin x terms.
✗ **(C)** Only took derivative of second term.
✗ **(D)** Sign error on cos derivative.

Question 28 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = e^x \cos x + e^x (-\sin x) = e^x(\cos x - \sin x)$. $f'(0) = e^0(1 - 0) = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Added sin term.
✗ **(C)** Sign error (1 – (−1)?).
✗ **(D)** Product rule error.

Question 29 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) −1/2

ANALYTICAL STEP-BY-STEP SOLUTION

Tangent slope $m = 2x = 2$. Normal slope $= -1/m = -1/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Tangent slope.
✗ **(B)** Negative tangent slope.
✗ **(D)** Reciprocal but positive.

Question 30 Analysis

★★★ MANDATORY

Correct Answer: (B) 1

ANALYTICAL STEP-BY-STEP SOLUTION

Simplify first: $f(x) = x - 1$ (for $x \neq -1$). $f'(x) = 1$. So $f'(2) = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluated f(2) instead of f'.
✗ **(C)** Squared the result.
✗ **(D)** Quotient rule arithmetic error.

Question 31 Analysis

★★★ MANDATORY

Correct Answer: (A) $x^4 e^x (5 + x)$

ANALYTICAL STEP-BY-STEP SOLUTION

Product Rule: $f' = 5x^4e^x + x^5e^x = x^4e^x(5 + x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Original function.
✗ **(C)** Sum rule error.
✗ **(D)** Incomplete product rule.

Question 32 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $-1/\pi$

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = (\cos x \cdot x - \sin x)/x^2$. At π: $(-1 \cdot \pi - 0)/\pi^2 = -\pi/\pi^2 = -1/\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Squared denominator error (forgot π in numerator).
✗ **(C)** $\sin \pi$ is 0, but $\cos \pi$ is -1.
✗ **(D)** Sign error.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $3^x \ln 3$

ANALYTICAL STEP-BY-STEP SOLUTION

Exponential rule: $d/dx (a^x) = a^x \ln a$. So $3^x \ln 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Missing $\ln 3$.
✗ **(C)** Power rule misapplied.
✗ **(D)** Just the constant.

Question 34 Analysis

★★★ MANDATORY

Correct Answer: (D) 1/e

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = 1/x. f'(e) = 1/e.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Derivative of constant?
✗ **(B)** f(e) value.
✗ **(C)** x value.

Question 35 Analysis

★★★ MANDATORY

Correct Answer: (D) (0, 1)

ANALYTICAL STEP-BY-STEP SOLUTION

Point is $(0, e^0) = (0, 1)$. Slope is $e^0 = 1$. Equation: $y - 1 = 1(x - 0) \Rightarrow y = x + 1$. The point of tangency is (0, 1).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** (1,0) gives $y = 2 \neq 0$.
✗ **(B)** (1,1) does not satisfy $y = x + 1$.
✗ **(C)** Does not pass through origin.

Question 36 Analysis

★★★ MANDATORY

Correct Answer: (C) x = −1

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = 2x + 2. Horizontal tangent requires f'(x) = 0: $2x = -2 \rightarrow x = -1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Positive root of f'.
✗ **(B)** Double the root.
✗ **(D)** Root of f.

Question 37 Analysis

★★★ MANDATORY

Correct Answer: (C) $-\csc^2 x$

ANALYTICAL STEP-BY-STEP SOLUTION

Standard trig derivative: $d/dx (\cot x) = -\csc^2 x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** tan derivative.
✗ **(B)** Negative secant.
✗ **(D)** Forgot negative sign.

Question 38 Analysis

★★★ MANDATORY

Correct Answer: (C) 3

ANALYTICAL STEP-BY-STEP SOLUTION

$f(x) = x^{(3/2)}$. $f'(x) = (3/2)x^{(1/2)} = (3/2)\sqrt{x}$. $f'(4) = (3/2)(2) = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot 3/2 factor.
✗ **(B)** Forgot 1/2 factor.
✗ **(D)** Forgot square root evaluation.

Question 39 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $x = \pi/2$

ANALYTICAL STEP-BY-STEP SOLUTION

Tangent slope $m = \cos(\pi/2) = 0$ (horizontal tangent $y = 1$). The normal line is perpendicular, so it is the vertical line $x = \pi/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Diagonal line.
✗ **(C)** x-axis.
✗ **(D)** This is the tangent line equation.

Question 40 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: f'(x) = 2e^(2x). At x = 0: $2e^0 = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot chain factor 2.

✗ **(C)** e^0 is 1, not e.

✗ **(D)** e^2 error.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1/x

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: (1/2x) · 2 = 1/x. Or use ln(2x) = ln 2 + ln x, whose derivative is 0 + 1/x = 1/x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot chain rule factor 2.

✗ **(C)** Derivative of constant ln 2 is 0.

✗ **(D)** Multiplied instead of divided.

Question 42 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $10x(x^2 + 1)^4$

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $5(x^2+1)^4 \cdot (2x) = 10x(x^2+1)^4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Unmultiplied form.

✗ **(C)** Outer derivative only.

✗ **(D)** Power subtraction error.

Question 43 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2x cos(x²)

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $\cos(x^2) \cdot d/dx(x^2) = 2x \cos(x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.
✗ **(B)** Outer derivative only.
✗ **(D)** Wrong outer derivative.

Question 44 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2e

ANALYTICAL STEP-BY-STEP SOLUTION

h'(x) = e^(x²) · 2x. At x = 1: $e^1 \cdot 2(1) = 2e$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** e^2 error.
✗ **(C)** Forgot 2x inner derivative.
✗ **(D)** $2e^2$ error.

Question 45 Analysis

★★★ MANDATORY

Correct Answer: (B) 3/(2√(3x+1))

ANALYTICAL STEP-BY-STEP SOLUTION

f(x) = (3x+1)^(1/2). f'(x) = (1/2)(3x+1)^(−1/2) · 3 = 3/(2√(3x+1)).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot 1/2 factor.
✗ **(C)** Forgot inner derivative 3.
✗ **(D)** Forgot both 1/2 and inner.

Question 46 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = 3(\sin x)^2 \cdot \cos x$. At $\pi/2$: $3(1)^2 \cdot \cos(\pi/2) = 3(1)(0) = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.

✗ **(B)** Forgot cos term.

✗ **(C)** Evaluated $3(1)^2$ without cos.

Question 47 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) –tan x

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $(1/\cos x) \cdot (-\sin x) = -\sin x / \cos x = -\tan x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** sec x error.

✗ **(B)** Negative secant error.

✗ **(C)** Sign error.

Question 48 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = e^(sin x) · cos x. At x = 0: $e^0 \cdot \cos 0 = 1 \cdot 1 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Thinks derivative is 0.

✗ **(B)** sin 1 error.

✗ **(D)** e^1 error.

Question 49 Analysis

★★★ MANDATORY

Correct Answer: (A) $2(x + \sqrt{x})(1 + 1/(2\sqrt{x}))$

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $2(x + \sqrt{x}) \cdot d/dx(x + \sqrt{x}) = 2(x + \sqrt{x})(1 + 1/(2\sqrt{x}))$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Expanded incorrectly.
✗ **(C)** Squared inner derivative.
✗ **(D)** Missing inner derivative.

Question 50 Analysis

★★★ MANDATORY

Correct Answer: (A) $-3x^2 \sin(x^3)$

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $-\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Wrong argument.
✗ **(C)** Sign error.
✗ **(D)** Missing inner derivative.

Question 51 Analysis

★★★ MANDATORY

Correct Answer: (C) 8

ANALYTICAL STEP-BY-STEP SOLUTION

$y' = 4(2x - 1)^3 \cdot 2 = 8(2x - 1)^3$. At $x = 1$: $8(1)^3 = 8$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error.
✗ **(B)** Multiplied by 2^4.
✗ **(D)** Forgot chain factor 2.

Question 52 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1/e

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = (1/ln x) · (1/x). At x = e: (1/1) · (1/e) = 1/e.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgot 1/x inner factor.

✗ **(C)** Same as A but unsimplified.

✗ **(D)** Forgot 1/ln x outer factor.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) e^(e^x) · e^x

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: e^(e^x) · d/dx(e^x) = e^(e^x) · e^x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Extra factor of e^x.

✗ **(C)** Outer derivative only.

✗ **(D)** Exponent addition rule misapplied.

Question 54 Analysis

★★★ MANDATORY

Correct Answer: (B) 0

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = (1/2)(x²+1)^(−1/2) · 2x = x/√(x²+1). At x = 0: 0/1 = 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot x in numerator.

✗ **(C)** Denominator is √1 = 1 ≠ 0.

✗ **(D)** Forgot 2x chain.

Question 55 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

Pythagorean identity: $\sin^2 x + \cos^2 x = 1$. The derivative of constant 1 is 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Derivative of $\cos^2 x$ only.

✗ **(B)** Derivative of $\sin^2 x$ only.

✗ **(C)** Function value, not derivative.

Question 56 Analysis

★★★ MANDATORY

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = 100(x^2 - 1)^{99} \cdot 2x$. At $x = 1$: the factor $(1^2 - 1) = 0$, so the whole product is 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculated $100 \cdot 2$.

✗ **(B)** Forgot $(x^2 - 1)$ becomes 0.

✗ **(C)** 1^{100} error.

Question 57 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\sec^2(\sqrt{x})/(2\sqrt{x})$

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $\sec^2(\sqrt{x}) \cdot (1/(2\sqrt{x})) = \sec^2(\sqrt{x})/(2\sqrt{x})$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Outer derivative only.

✗ **(B)** Derivative of secant.

✗ **(C)** Inner derivative only.

Question 58 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) x = 0

ANALYTICAL STEP-BY-STEP SOLUTION

f'(x) = e^(−x²) · (−2x). Since e^(−x²) > 0 for all real x, f'(x) = 0 requires −2x = 0 → x = 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Same error.
✗ **(B)** f'(0) = 0 exists.
✗ **(D)** e^(−1) ≠ 0.

Question 59 Analysis

★★★ MANDATORY

Correct Answer: (A) 2/x

ANALYTICAL STEP-BY-STEP SOLUTION

(1/x²) · 2x = 2/x. Or ln(x²) = 2 ln x, whose derivative is 2/x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Inner derivative only.
✗ **(C)** Forgot chain rule.
✗ **(D)** Forgot 2x numerator.

Question 60 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −x/y

ANALYTICAL STEP-BY-STEP SOLUTION

Differentiate implicitly: 2x + 2y y' = 0 → 2y y' = −2x → y' = −x/y.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error.
✗ **(C)** Inverted quotient sign error.
✗ **(D)** Inverted quotient.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −y/x

ANALYTICAL STEP-BY-STEP SOLUTION

Product Rule: $1 \cdot y + x \cdot y' = 0 \rightarrow x\,y' = -y \rightarrow y' = -y/x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error.
✗ **(C)** Inverted quotient sign error.
✗ **(D)** Inverted quotient.

Question 62 Analysis

★★★ MANDATORY

Correct Answer: (C) −3/4

ANALYTICAL STEP-BY-STEP SOLUTION

$y' = -x/y = -6/8 = -3/4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverted positive error.
✗ **(B)** Inverted quotient.
✗ **(D)** Positive slope error.

Question 63 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) cos(xy)(y + xy') = 1

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $\cos(xy) \cdot d/dx(xy) = 1 \rightarrow \cos(xy)(y + xy') = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot product rule inside.
✗ **(B)** Forgot cos on second term.
✗ **(C)** Forgot chain rule.

Question 64 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) −1

ANALYTICAL STEP-BY-STEP SOLUTION

$3x^2 + 3y^2\,y' = 6(y + xy')$. Divide by 3: $x^2 + y^2\,y' = 2y + 2xy'$. At (3, 3): $9 + 9y' = 6 + 6y' \rightarrow 3y' = -3 \rightarrow y' = -1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.

✗ **(B)** Denominator is $9 - 6 = 3 \neq 0$.

✗ **(C)** Algebraic slip.

Question 65 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $x^2 + y^2 = 100$

ANALYTICAL STEP-BY-STEP SOLUTION

By the Pythagorean theorem for the right triangle formed by the ladder: $x^2 + y^2 = 10^2 = 100$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Area relation.

✗ **(C)** Perimeter relation.

✗ **(D)** Hyperbola equation.

Question 66 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 12π cm²/s

ANALYTICAL STEP-BY-STEP SOLUTION

$A = \pi r^2$. Differentiate with respect to t: $dA/dt = 2\pi r\,(dr/dt)$. When $r = 3$ and $dr/dt = 2$: $2\pi(3)(2) = 12\pi$ cm²/s.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot 2 in power rule.

✗ **(B)** Forgot radius factor.

✗ **(C)** Forgot dr/dt factor.

Question 67 Analysis

★★★ MANDATORY

Correct Answer: (B) x/y

ANALYTICAL STEP-BY-STEP SOLUTION

$2x - 2y\,y' = 0 \rightarrow 2y\,y' = 2x \rightarrow y' = x/y$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error (circle case).
✗ **(C)** Inverted sign error.
✗ **(D)** Inverted quotient.

Question 68 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) f'(g(t)) g'(t)

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: $dy/dt = (dy/du) \cdot (du/dt) = f'(u) \cdot g'(t) = f'(g(t))\,g'(t)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Composition error.
✗ **(C)** Equivalent raw form but B is explicit composite notation.
✗ **(D)** Inverted composition.

Question 69 Analysis

★★★ MANDATORY

Correct Answer: (C) y = 0

ANALYTICAL STEP-BY-STEP SOLUTION

$dy/dx = -x/y$. A vertical tangent occurs where the denominator is 0 (and numerator $\neq 0$), so $y = 0$ (at points $(\pm 5, 0)$).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Horizontal tangent condition.
✗ **(B)** $x = 5$ only gives one point.
✗ **(D)** $y = 5$ gives horizontal tangent.

Question 70 Analysis

★★ IMPORTANT

Correct Answer: (A) −2/3

ANALYTICAL STEP-BY-STEP SOLUTION

Differentiate with respect to x: 2 + 3y' = 0 → y' = −2/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Positive slope.
✗ **(C)** Inverted positive slope.
✗ **(D)** Inverted slope.

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) e^(xy)(y + xy') = 1 + y'

ANALYTICAL STEP-BY-STEP SOLUTION

Chain Rule: e^(xy) · d/dx(xy) = d/dx(x + y) → e^(xy)(y + xy') = 1 + y'.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgot x term.
✗ **(B)** Forgot product rule inside exponent.
✗ **(C)** Forgot y term.

Question 72 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) −cot t

ANALYTICAL STEP-BY-STEP SOLUTION

Parametric derivative: dy/dx = (dy/dt)/(dx/dt) = (cos t)/(−sin t) = −cot t.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverted tangent sign error.
✗ **(C)** Inverted tangent.
✗ **(D)** Sign error.

Question 73 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $4\pi r^2 k$

ANALYTICAL STEP-BY-STEP SOLUTION

$dV/dt = 4\pi r^2 (dr/dt) = 4\pi r^2 k$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Volume derivative missing 3 cancellation.
✗ **(B)** Area formula missing 4.
✗ **(D)** Missing r.

Question 74 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $-1/(4y^3)$

ANALYTICAL STEP-BY-STEP SOLUTION

$2y\,y' = 1 \rightarrow y' = 1/(2y)$. $y'' = -2y' / (2y)^2 = -2(1/2y) / 4y^2 = -1/(4y^3)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Missing factor of 4.
✗ **(C)** Sign error.
✗ **(D)** First derivative only.

Question 75 Analysis

★★★ MANDATORY

Correct Answer: (D) 1

ANALYTICAL STEP-BY-STEP SOLUTION

$y' = -y/x = -2/2 = -1$. The normal slope is perpendicular: $m_{norm} = -1/(-1) = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Double slope.
✗ **(B)** Tangent slope.
✗ **(C)** Negative double.

Question 76 Analysis

★★ IMPORTANT

Correct Answer: (A) Increasing

ANALYTICAL STEP-BY-STEP SOLUTION

A strictly positive first derivative implies the function f is increasing on that interval.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Requires f'' < 0.
✗ **(C)** Requires f'' > 0.
✗ **(D)** Requires f' < 0.

Question 77 Analysis

★★ IMPORTANT

Correct Answer: (C) Concave Down

ANALYTICAL STEP-BY-STEP SOLUTION

A negative second derivative implies the graph of f is concave down on that interval.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** f' < 0.
✗ **(B)** f' > 0.
✗ **(D)** f'' > 0.

Question 78 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) f''(x) = 0 or undefined, and concavity changes sign

ANALYTICAL STEP-BY-STEP SOLUTION

An inflection point requires a change in concavity, which occurs where f'' changes sign.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Extrema of slope.
✗ **(B)** Critical point condition.
✗ **(C)** x-intercept condition.

Question 79 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) Local Minimum at c

ANALYTICAL STEP-BY-STEP SOLUTION

Second Derivative Test: f'(c) = 0 and f''(c) > 0 (concave up) guarantees a local minimum at x = c.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Requires f'' to change sign.

✗ **(B)** Test is conclusive.

✗ **(C)** Requires f''(c) < 0.

Question 80 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) f' crosses the x-axis from positive to negative

ANALYTICAL STEP-BY-STEP SOLUTION

By the First Derivative Test, f has a local maximum where f' changes sign from positive to negative.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Point of inflection condition.

✗ **(B)** Point of inflection condition.

✗ **(D)** Local minimum condition.

Question 81 Analysis

★★★ MANDATORY

Correct Answer: (D) 0

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = 4x^3$, $f''(x) = 12x^2$. At $x = 0$: $12(0)^2 = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Second derivative coefficient without substitution.

✗ **(B)** First derivative coefficient.

✗ **(C)** Third derivative.

Question 82 Analysis

★★ IMPORTANT

Correct Answer: (C) Position s(t)

ANALYTICAL STEP-BY-STEP SOLUTION

Instantaneous velocity is the derivative of position: v(t) = s'(t).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Acceleration is v'(t).
✗ **(B)** Time is the independent variable.
✗ **(D)** Speed is the magnitude of velocity.

Question 83 Analysis

★★ IMPORTANT

Correct Answer: (D) Velocity v(t)

ANALYTICAL STEP-BY-STEP SOLUTION

Instantaneous acceleration is the rate of change of velocity: a(t) = v'(t) = s''(t).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** First derivative of position is velocity.
✗ **(B)** Displacement is net position change.
✗ **(C)** Speed is a scalar magnitude.

Question 84 Analysis

★★★ MANDATORY

Correct Answer: (B) 0, 2, −1

ANALYTICAL STEP-BY-STEP SOLUTION

Critical numbers occur where f'(x) = 0: x = 0, x = 2, x = −1.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign errors on roots.
✗ **(C)** Missing −1.
✗ **(D)** Missing 0.

Question 85 Analysis

★★★ MANDATORY

Correct Answer: (D) x = 1

ANALYTICAL STEP-BY-STEP SOLUTION

Set f''(x) = 0: 6x − 6 = 0 → x = 1. Since f'' is linear, it changes sign across x = 1, giving an inflection point.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** f''(−1) = −12 ≠ 0.

✗ **(B)** f''(0) = −6 ≠ 0.

✗ **(C)** f''(6) = 30 ≠ 0.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 7

ANALYTICAL STEP-BY-STEP SOLUTION

Product Rule: (f · g)'(2) = f'(2)g(2) + f(2)g'(2) = (5)(−1) + (3)(4) = −5 + 12 = 7.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplied derivatives.

✗ **(B)** Multiplied derivatives only: (5)(4) = 20.

✗ **(C)** Arithmetic error: 5 + 12 = 17.

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) x = 3

ANALYTICAL STEP-BY-STEP SOLUTION

f has a sharp corner (cusp) at x = 3. The left-hand derivative is −1 while the right-hand derivative is +1. Since −1 ≠ 1, f is not differentiable at x = 3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error.

✗ **(B)** Absolute value has sharp corners where the inside vanishes.

✗ **(C)** Corner is at x = 3, not 0.

Question 88 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) x = 1 and x = −1

ANALYTICAL STEP-BY-STEP SOLUTION

Quotient Rule: $f'(x) = [1\cdot(x^2 + 1) - x\cdot(2x)] / (x^2 + 1)^2 = (1 - x^2) / (x^2 + 1)^2$. Horizontal tangent requires $f'(x) = 0$: $1 - x^2 = 0 \rightarrow x = \pm 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** $f'(x)$ has real zeros at ±1.

✗ **(C)** Misses negative root $x = -1$.

✗ **(D)** At $x = 0$, $f'(0) = 1 \neq 0$.

Question 89 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1

ANALYTICAL STEP-BY-STEP SOLUTION

$f'(x) = \sec x \tan x$. Product Rule: $f''(x) = (\sec x \tan x) \tan x + \sec x (\sec^2 x) = \sec x \tan^2 x + \sec^3 x$. At $x = 0$: $\sec 0 = 1$, $\tan 0 = 0$. $f''(0) = (1)(0) + (1)^3 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** $\tan 0$ is 0, but $\sec^3 0$ is 1.

✗ **(C)** Double addition error.

✗ **(D)** Sign error.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2

ANALYTICAL STEP-BY-STEP SOLUTION

Average rate = $[f(4) - f(0)] / (4 - 0) = [8 - 0] / 4 = 2$. Instantaneous rate $f'(c) = 2c - 2$. Set $2c - 2 = 2 \rightarrow 2c = 4 \rightarrow c = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Midpoint error.

✗ **(C)** Arithmetic error in average rate.

✗ **(D)** Endpoint.

Question 91 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $[f(3) - f(1)]/(3 - 1)$. $f(3) = 9 - 12 + 5 = 2$. $f(1) = 1 - 4 + 5 = 2$. $(2 - 2)/2 = \mathbf{0}$.

(b) $f'(x) = \lim_{h\to 0} [((x+h)^2 - 4(x+h) + 5) - (x^2 - 4x + 5)]/h$

$= \lim [x^2 + 2xh + h^2 - 4x - 4h + 5 - x^2 + 4x - 5]/h$

$= \lim [2xh + h^2 - 4h]/h = \lim (2x + h - 4) = \mathbf{2x - 4}$.

(c) $f(2) = 4 - 8 + 5 = 1$. Point $(2, 1)$. $f'(2) = 2(2) - 4 = 0$. Tangent: $\mathbf{y - 1 = 0(x - 2)} \Rightarrow \mathbf{y = 1}$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct setup and evaluation of average rate = 0	2 pts
Part (b): Correct expansion and simplification of difference quotient	2 pts
Part (b): Correct limit evaluation yielding $f'(x) = 2x - 4$	2 pts
Part (c): Correct point (2, 1), slope m = 0, and equation y = 1	3 pts

Question 92 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $f'(x) = (2x)(e^x) + (x^2)(e^x) = \mathbf{e^x(2x + x^2)}$.

(b) $g'(x) = [(1)(x - 1) - (x + 1)(1)] / (x - 1)^2 = (x - 1 - x - 1)/(x - 1)^2 = \mathbf{-2/(x - 1)^2}$.

(c) $f'(0) = e^0(0) = \mathbf{0}$. $g'(2) = -2/(1)^2 = \mathbf{-2}$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct application of Product Rule yielding $e^x(2x + x^2)$	3 pts
Part (b): Correct application of Quotient Rule yielding $-2/(x - 1)^2$	3 pts
Part (c): Correct evaluations $f'(0) = 0$ and $g'(2) = -2$	3 pts

Question 93 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $h'(x) = \cos(e^{2x}) \cdot \frac{d}{dx}(e^{2x}) = \cos(e^{2x}) \cdot e^{2x} \cdot 2 =$ **$2e^{2x}\cos(e^{2x})$**.

(b) $k'(x) = (1/\cos x) \cdot (-\sin x) =$ **$-\tan x$**.

(c) $h'(0) = 2e^{0}\cos(e^{0}) =$ **$2\cos 1$**. $k'(\pi/4) = -\tan(\pi/4) =$ **-1**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct double chain rule application yielding $2e^{2x}\cos(e^{2x})$	3 pts
Part (b): Correct chain rule yielding $-\tan x$	3 pts
Part (c): Correct evaluations $h'(0) = 2\cos 1$ and $k'(\pi/4) = -1$	3 pts

Question 94 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) Diff wrt x: $2x + (1\cdot y + x\cdot y') + 2y\cdot y' = 0$.

$2x + y + y'(x + 2y) = 0 \Rightarrow y'(x + 2y) = -(2x + y) \Rightarrow$ **$dy/dx = -(2x + y)/(x + 2y)$**.

(b) Set $x = 1$: $1 + y + y^2 = 3 \Rightarrow y^2 + y - 2 = 0 \Rightarrow (y + 2)(y - 1) = 0$. Points: **(1, 1) and (1, −2)**.

(c) At (1, 1): $dy/dx = -(2 + 1)/(1 + 2) = -1$. Eq: **$y - 1 = -1(x - 1) \Rightarrow y = -x + 2$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct implicit differentiation and solving for dy/dx	3 pts
Part (b): Solving quadratic for y, finding (1, 1) and (1, −2)	2 pts
Part (c): Correct slope m = −1 and tangent equation y = −x + 2	4 pts

Question 95 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) Pythagorean: $\mathbf{x^2 + y^2 = 100}$.

(b) Diff wrt t: $2x(dx/dt) + 2y(dy/dt) = 0 \Rightarrow x(dx/dt) + y(dy/dt) = 0$.

When $x = 6$, $y = \sqrt{(100 - 36)} = 8$. $dx/dt = 2$.

$6(2) + 8(dy/dt) = 0 \Rightarrow 12 + 8(dy/dt) = 0 \Rightarrow dy/dt = -12/8 =$ **-1.5 ft/sec**.

(c) Since $dy/dt < 0$, the top of the ladder is **moving down** at 1.5 ft/sec.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct geometric relation $x^2 + y^2 = 100$	2 pts
Part (b): Correct differentiation and substitution	2 pts
Part (b): Correct solution $dy/dt = -1.5$ ft/sec	2 pts
Part (c): Correct negative sign interpretation (moving down)	3 pts

Question 96 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $f'(x) = 4x^3 - 12x^2$. $f''(x) = 12x^2 - 24x$.

(b) Set $f''(x) = 0$: $12x(x - 2) = 0 \Rightarrow x = 0, x = 2$. Check signs: f'' changes sign at both. Points of inflection at **$x = 0, 2$**.

(c) $f''(x) > 0$ when $x < 0$ or $x > 2$. Concave up on **$(-\infty, 0) \cup (2, \infty)$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct first and second derivatives	2 pts
Part (b): Finding zeros of f'' and verifying sign changes at $x = 0, 2$	3 pts
Part (c): Correct open intervals for concave up $(-\infty, 0) \cup (2, \infty)$	4 pts

Question 97 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) f increases where f' > 0: **(−2, 1) and (4, ∞)**.

(b) Local max where f' changes from + to −: **x = 1**.

(c) By the Fundamental Theorem: $f(2) = f(0) + \int_0^2 f'(x)\,dx = 5 + (-3) = \mathbf{2}$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Identifying intervals where f' > 0	2 pts
Part (b): Identifying sign change + to − for local maximum at x = 1	3 pts
Part (c): Using Net Change Theorem correctly: f(2) = 5 − 3 = 2	4 pts

Question 98 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $dx/dt = 2t$, $dy/dt = 3t^2 - 3$. $dy/dx = (dy/dt)/(dx/dt) = \mathbf{(3t^2 - 3)/(2t)}$.

(b) At $t = 2$: $(3(4) - 3)/(4) = 9/4 = \mathbf{2.25}$.

(c) Horizontal tangent when $dy/dt = 0$ (and $dx/dt \neq 0$). $3t^2 - 3 = 0 \Rightarrow t = \pm 1$. At $t = \pm 1$, $dx/dt = \pm 2 \neq 0$. Values: **$t = 1, -1$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct parametric derivative formula $dy/dx = (3t^2 - 3)/(2t)$	**3 pts**
Part (b): Correct evaluation at $t = 2$ yielding $9/4$	**2 pts**
Part (c): Solving $dy/dt = 0$, verifying $dx/dt \neq 0$, finding $t = \pm 1$	**4 pts**

Question 99 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $y - 2 = -3(x - 4) \Rightarrow \mathbf{y = -3x + 14}$.

(b) $f(4.1) \approx -3(4.1) + 14 = -12.3 + 14 = \mathbf{1.7}$.

(c) Since $f''(x) < 0$, the graph is **concave down**. The tangent line lies above the curve, so the linear approximation is an **overestimate**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct tangent line equation $y - 2 = -3(x - 4)$	2 pts
Part (b): Correct substitution and approximation $f(4.1) \approx 1.7$	3 pts
Part (c): Correctly linking concavity ($f'' < 0$) to overestimate	4 pts

Question 100 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $f'(x) = 1 + 2\cos x$. $f''(x) = -2\sin x$.

(b) $f'(x) = 0 \Rightarrow \cos x = -1/2$. In $[0, 2\pi]$: **$x = 2\pi/3, 4\pi/3$**.

(c) $f''(2\pi/3) = -2\sin(2\pi/3) = -2(\sqrt{3}/2) = -\sqrt{3} < 0 \Rightarrow$ **Local Maximum**.

$f''(4\pi/3) = -2\sin(4\pi/3) = -2(-\sqrt{3}/2) = +\sqrt{3} > 0 \Rightarrow$ **Local Minimum**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct first and second derivatives	2 pts
Part (b): Solving $1 + 2\cos x = 0$ for $x = 2\pi/3, 4\pi/3$	3 pts
Part (c): Correct application of 2nd derivative test and classification	4 pts

Essential Unit 2 Derivative Rules & Formulas

- **Definition of Derivative:** $f'(x) = \lim_{h\to 0} [f(x+h) - f(x)]/h = \lim_{x\to c} [f(x) - f(c)]/(x - c)$.
- **Power Rule:** $d/dx (x^n) = n x^{n-1}$.
- **Product Rule:** $(u \cdot v)' = u'v + uv'$.
- **Quotient Rule:** $(u / v)' = (u'v - uv') / v^2$.
- **Chain Rule:** $d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$.
- **Trig Derivatives:** $(\sin x)' = \cos x$ | $(\cos x)' = -\sin x$ | $(\tan x)' = \sec^2 x$ | $(\sec x)' = \sec x \tan x$ | $(\cot x)' = -\csc^2 x$ | $(\csc x)' = -\csc x \cot x$.
- **Exponential & Logarithms:** $(e^x)' = e^x$ | $(a^x)' = a^x \ln a$ | $(\ln x)' = 1/x$ | $(\log_a x)' = 1/(x \ln a)$.
- **Implicit Differentiation:** Differentiate both sides wrt x, apply chain rule (dy/dx) on y terms, and isolate dy/dx.

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